RAMCO INSTITUTE OF TECHNOLOGY

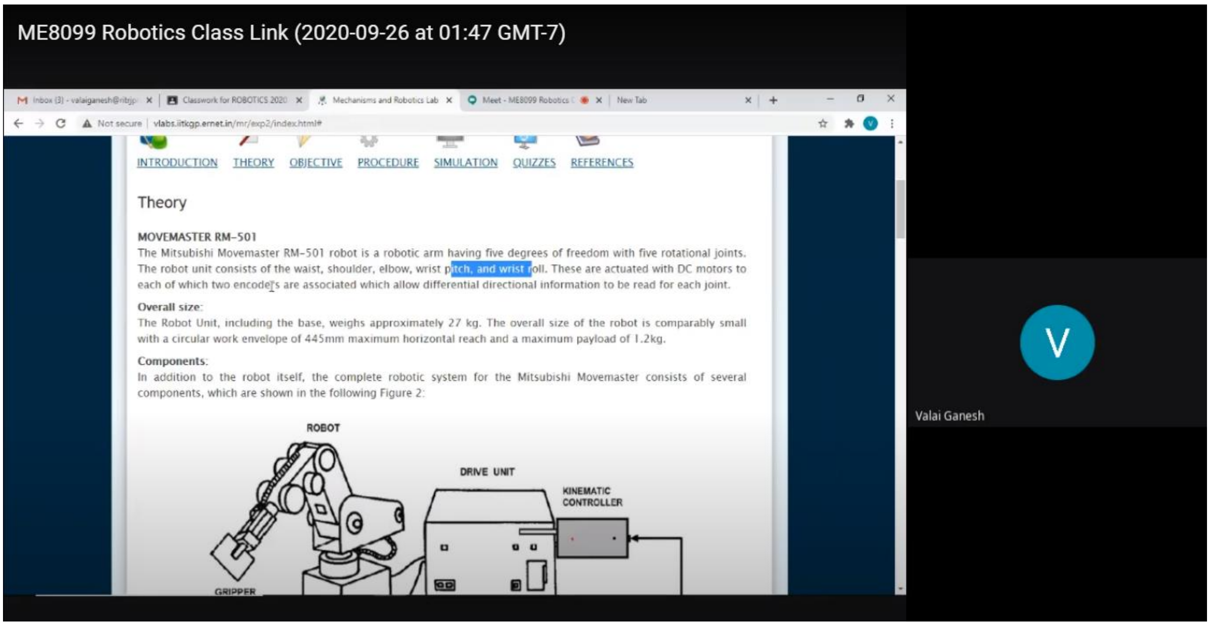
Department of Mechanical Engineering

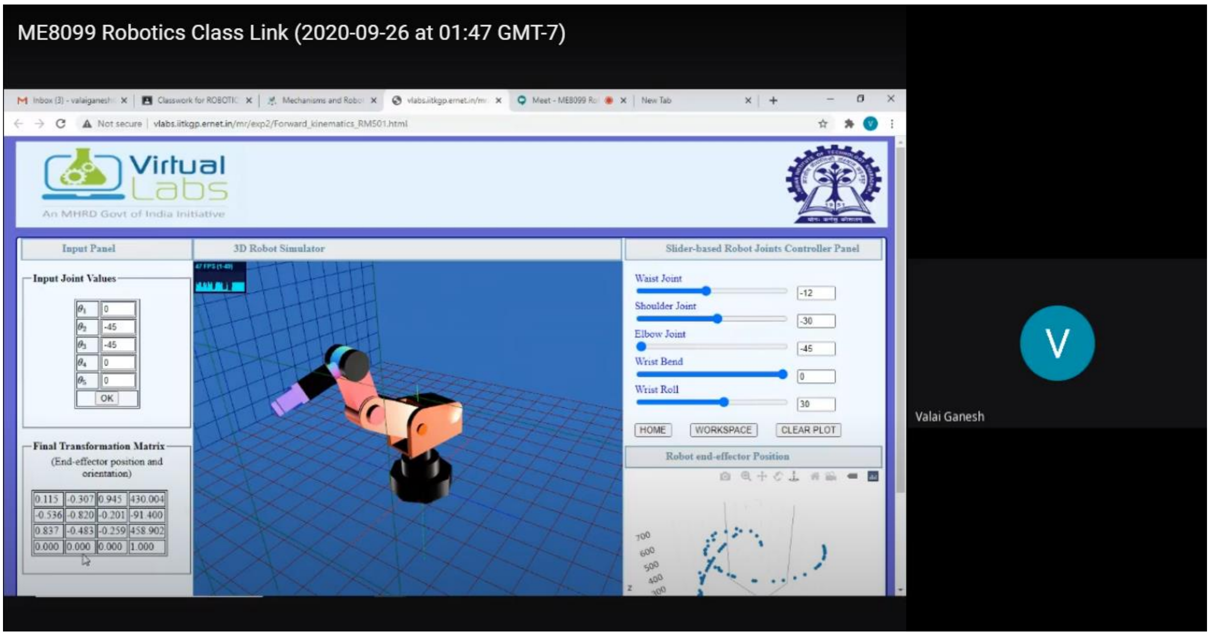
Academic Year: 2020-2021 (Odd Semester)

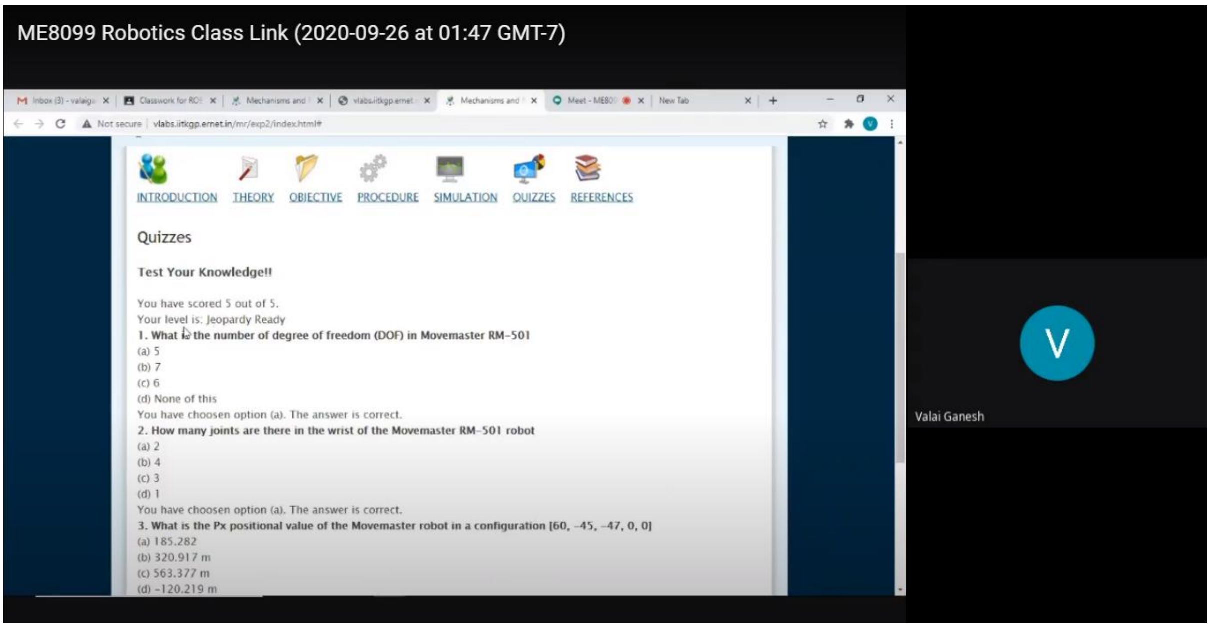
Innovative Practices- Virtual Laboratory Session Handling Details

Name of the Faculty	:	Mr. S. Valai Ganesh
Subject Code and Title	:	ME8099 Robotics
Year and Branch	:	IV year Mechanical Engineering
Date	:	26.09.2020
Topics Covered	:	Robot Kinematics
Name of the Laboratory	:	Mechanism and Robotics Laboratory
Website Link	:	http://vlabs.iitkgp.ernet.in/mr/#

The PUMA 560 is an industrial robot arm with six degrees of freedom and all rotational joints. In this experiment at first a brief theory about PUMA 560 robot is presented in the theory section. The theory for mathematical computations was obtained from a wide variety of sources encompassing books, papers and internet. In simulation section a virtual model is developed in JavaScript program which is used to investigate the forward kinematics problem.




ME8099 Robotics Class Link (2020-09-26 at 01:47 GMT-7)
vlabs.iitkgp.ernet.in/mr/exp2/Forward_kinematics_RM501.html
Virtual Labs
An MHRD Govt of India Initiative
Input Panel
3D Robot Simulator
Slider-based Robot Joints Controller Panel
Input Joint Values
OK
Waist Joint
Shoulder Joint
Elbow Joint
Wrist Bend
Wrist Roll
HOME
WORKSPACE
CLEAR PLOT
Robot end-effector Position
Final Transformation Matrix
(End-effector position and orientation)
0.115 -0.307 0.945 430.004
-0.536 -0.820 -0.201 -91.400
0.837 -0.483 -0.259 458.902
0.000 0.000 0.000 1.000
Valai Ganesh


ME8099 Robotics Class Link (2020-09-26 at 01:47 GMT-7)
vlabs.iitkgp.ernet.in/mr/exp2/index.html#
INTRODUCTION
THEORY
OBJECTIVE
PROCEDURE
SIMULATION
QUIZZES
REFERENCES
Quizzes
Test Your Knowledge!!
You have scored 5 out of 5.
Your level is: Jeopardy Ready
1. What is the number of degree of freedom (DOF) in Movemaster RM-501
(a) 5
(b) 7
(c) 6
(d) None of this
You have choosen option (a). The answer is correct.
2. How many joints are there in the wrist of the Movemaster RM-501 robot
(a) 2
(b) 4
(c) 3
(d) 1
You have choosen option (a). The answer is correct.
3. What is the Px positional value of the Movemaster robot in a configuration [60, -45, -47, 0, 0]
(a) 185.282
(b) 320.917 m
(c) 563.377 m
(d) -120.219 m
Valai Ganesh